

VIVEK KUMAR

Senior Technical Lead · Software Architect Track

viveksingh0143@gmail.com | +91-9910896849 | Ghaziabad, Uttar Pradesh, India
linkedin.com/in/viveksingh0143 | mytechbytes.in

PROFESSIONAL SUMMARY

Senior Technical Lead on a Software Architect track with 14+ years of experience designing and delivering large-scale distributed systems — breaking monoliths into microservices, shaping event-driven architectures, and embedding security and scale from the ground up across airline, media, and publishing domains.

CORE COMPETENCIES

Architecture & Design	Microservices · Event-Driven Architecture · Distributed Systems · Domain-Driven Design · API Gateway · Sidecar Pattern · System Design & Scalability
Backend	Java 17 · Spring Boot · Spring Cloud · Spring MVC · Hibernate · Struts 2 · REST APIs · .NET 10
Frontend	React.js · AngularJS · Vue.js · Micro-frontends · Module Federation · Webpack
Messaging & Data	Apache Kafka · Solace PubSub+ · Couchbase · Oracle · MySQL · Apache Solr · Azure Blob Storage
AI & Emerging	Elixir/Phoenix LiveView · Python/FastAPI · LLM Integration · RAG · Agentic Workflows
Platform & Practice	Token-based Auth · Azure AD · JWT · CI/CD · Jenkins · Docker · Git · VAPT / PT Remediation · Agile / Scrum · Mentoring

PROFESSIONAL EXPERIENCE

Senior Technical Lead — Software Architect Track

Aug 2022 – Present

Coforge · Client: Emirates (Dnata Travel / Emirates Group) · Greater Noida | Promoted from Technology Specialist

- Lead technical direction for the ResConnect airline booking platform at Emirates, guiding a team of 7+ developers and QA engineers through system design and delivery.
- Driving incremental monolith decomposition into independently deployable microservices — designed Booking and Ancillary (Bags) services end-to-end using Java 17 and Spring Boot.
- Architected the API gateway and centralised token-validation service (Spring Cloud Gateway, Sidecar pattern, Azure AD / JWT) across all services.
- Built event-driven screen-recording control via Solace PubSub+ — auto-pauses on payment entry for PCI compliance; zero-downtime TIBCO → Wasola (SOAP → REST) migration recognised by the client.
- Designed RBAC with role-location-permission mapping in Couchbase; built microfrontend POC (Module Federation) integrating React with Angular for independent UI delivery.
- [In Progress] Image offload for Dnata Travel — SQL Server BLOBs to Azure Blob Storage via .NET 10 REST API with source_flag-based zero-downtime cutover.
- Remediated VAPT / penetration-test findings; mentored engineers through design and code reviews; adopted AI tooling for test generation and legacy-code comprehension.

Stack: Java 17 · Spring Boot · Spring Cloud Gateway · Solace PubSub+ · Couchbase · React.js · AngularJS · Azure AD · JWT · .NET 10 · Dapper · Azure Blob Storage · CI/CD · Docker

Senior Software Engineer

Sep 2010 – Aug 2022

Hindustan Times

- Designed and delivered large-scale content publishing platforms, owning database design, backend services, and search — supporting millions of articles and images across multiple publications.
- Built scalable REST APIs and backend systems with Spring, Struts 2, and Hibernate; designed Apache Solr search with custom indexing strategies and query optimisation for high-volume news content.
- Engineered an automated content-aggregation platform (Selenium + HTMLUnit) crawling 200+ sources, with ingestion, transformation, and FTP/RSS delivery pipelines.
- Developed a mobile content backend (NewZZa) serving articles and media to iOS/Android apps with Solr-indexed retrieval and filtering.
- Built and maintained CMS web interfaces using Vue.js and AngularJS; optimised MySQL queries and indexing for high-volume transactional processing.
- Led a small team ensuring code quality, system stability, and on-time delivery across multiple concurrent projects.

Stack: Java · Spring · Struts 2 · Hibernate · MySQL · Apache Solr · Selenium · HTMLUnit · Vue.js · AngularJS · RSS · FTP

Executive — R&D

Jun 2007 – Jul 2010

Thomson Digital · Thomson Press (India) Ltd.

- Designed and built XML-driven publishing automation workflows using the Adobe InDesign JavaScript SDK, achieving 80–90% automated page typesetting for publishers including Elsevier and Pearson.
- Built XML transformation pipelines (Java, XSLT, XPath) against publisher-specific DTDs, plus QA automation scripts for XML validation and content-quality checks.

Stack: Java · XML · XSLT · XPath · Adobe InDesign JS SDK

KEY ACHIEVEMENTS

- **Zero-Downtime Migration:** Migrated stateless services from TIBCO (SOAP) to Wasola (REST) using a microservices + feature-flag strategy with no production impact — formally recognised by the Emirates/Coforge client.
- **Monolith → Microservices:** Led the step-by-step decomposition of the ResConnect booking platform, shaping the service design that the entire modernised platform is now built on.
- **PCI-Safe Screen Recording:** Designed an event-driven Kafka/Solace mechanism that automatically pauses agent screen recording during payment entry, ensuring zero card-data capture.
- **AI-Assisted Delivery:** Adopted AI tooling for unit-test generation and legacy-code comprehension, measurably improving team throughput on complex legacy modules.
- **Independent SaaS Products:** Self-directed: built and deployed QuoteAssist — an AI-powered multi-tenant lead-to-quote platform (Elixir/Phoenix LiveView + Python/FastAPI) on production OCI infrastructure.

EDUCATION

B.Sc. in Computer Science

Sikkim Manipal University

Diploma in Information Technology

Board of Technical Education, Uttar Pradesh

VIVEK KUMAR — PROJECT PORTFOLIO

Senior Technical Lead · mytechbytes.in · viveksingh0143@gmail.com

AIRLINE PLATFORM — RESCONNECT (COFORGE · EMIRATES)

Microservices Transformation — Booking & Ancillary

2024 – 2026

Senior Technical Lead / Architect Track · Coforge

Incremental decomposition of the legacy ResConnect monolith into independently deployable microservices across four active modules, with hands-on architecture and full-stack implementation.

- Ancillary Services (Bags): end-to-end UI and backend delivery in Java 17 / Spring Boot; Oracle + Couchbase data layer.
- TIBCO → Wasola Migration: extracted SOAP-based transformation logic into a dedicated REST microservice, enabling clean monolith removal without a big-bang rewrite.
- Data Layer Optimisation: targeted query and throughput improvements across Oracle (transactional) and Couchbase (distributed) stores.
- Monolith Modularisation: split monolith into independent frontend and backend repositories (Jan 2024 – Aug 2024), retaining a single deployable as the migration foundation.

Stack: Java 17 · Spring Boot · React.js · AngularJS · Oracle · Couchbase · Solace · REST APIs

Authentication — Centralised Token Validation & Sidecar Pattern

2024 – 2025

Software Architect · Coforge

Centralised access-token validation service securing all microservices via the Sidecar pattern, eliminating duplicated auth logic and providing consistent enforcement across the platform.

- Built a dedicated Spring Cloud Gateway filter chain intercepting every inbound request and validating JWTs against Azure AD's JWKS introspection endpoint.
- Sidecar deployed per pod (Kubernetes): valid tokens propagate to the app container; invalid tokens return 401 upstream-call-free.
- Claims extraction and role enforcement via Spring Security / SecurityContextHolder, integrated with Azure AD (Spring Cloud Azure AD).

Stack: Java 17 · Spring Cloud Gateway · Azure AD · JWT · Spring Security · Sidecar Pattern · Kubernetes

Microfrontend POC — Module Federation

May 2024 – Aug 2024

Software Architect · Coforge

Proof of concept modernising the agent booking UI into independently releasable microfrontend modules, integrating React with the existing AngularJS host.

- Implemented Webpack Module Federation to expose React-based feature modules consumed by the AngularJS shell application.
- Demonstrated independent build, deploy, and versioning per UI domain without disrupting the host app.

Stack: React.js · AngularJS · Webpack · Module Federation

Access Management — RBAC Admin UI

Jan 2024 – Aug 2024

Senior Technical Lead · Coforge

Role-based access control across the airline platform: role-location-permission mapping in Couchbase with a React.js admin interface for managing roles and permissions.

- Designed the Couchbase document schema for role → location → permission hierarchies supporting multi-location airline workflows.
- Built the React.js admin UI for CRUD management of roles and permissions, backed by Spring Boot REST APIs.

Stack: Spring Boot · React.js · Couchbase · RBAC

SQL Server → Azure Blob Storage Image Migration (Dnata Travel) [In Progress] Jun 2026 – Present

Software Architect · Coforge

Zero-downtime migration of binary image assets stored in SQL Server BLOB columns to Azure Blob Storage, with dual-read compatibility during cutover.

- Designed a source_flag-based migration pipeline: legacy rows read from SQL Server while new writes go directly to Azure Blob Storage.
- Built a .NET 10 REST API (Dapper) exposing SAS URL-based image retrieval; API handles both legacy and migrated images transparently.
- Zero-downtime cutover strategy: both stores active in parallel until validation passes, then legacy reads retired.

Stack: .NET 10 · Dapper · Azure Blob Storage · SAS URLs · MS SQL Server · REST API

CONTENT PLATFORMS (HINDUSTAN TIMES)

UI-Based Auto Crawling & Scraping Platform

2017 – 2021

Senior Software Engineer · Hindustan Times

End-to-end content automation platform — define scraping rules, crawl 200+ sources, template articles, and deliver via FTP and RSS through a unified workflow.

- Configuration management UI (Vue.js / Quasar / Spring Boot): per-site CSS selector and XPath rules.
- Crawling & scraping engine using Selenium and HTMLUnit for dynamic and static site extraction.
- Article templating to structured content, then automated FTP and RSS delivery to clients.

Stack: Java · Spring Boot · Vue.js · Quasar · Selenium · HTMLUnit · FTP · RSS

HTS Portal — Large-Scale Content & Search Platform

2012 – 2014

Software Engineer · Hindustan Times

Content platform managing millions of articles and images across multiple publications, with high-volume FTP ingestion and Apache Solr-powered search.

- Spring DI / Struts 2 MVC / Hibernate ORM application with Bootstrap web interface.
- FTP ingestion pipeline fetching articles and images and storing with structured metadata.
- Apache Solr indexing with custom filtering and query optimisation for high-performance retrieval at scale.
- MySQL query and index tuning for high-volume transactional content processing.

Stack: Java · Spring · Struts 2 · Hibernate · MySQL · Apache Solr · Bootstrap · FTP

NewZZa — Mobile Content Backend Feb 2015 – Dec 2015

Senior Software Engineer · Hindustan Times

REST backend serving articles and media to iOS and Android apps, with Solr-indexed content retrieval and category filtering.

Stack: Spring MVC · Hibernate · Apache Solr · MySQL

NMS — RSS Delivery & Splitter

Mar 2016 – Jul 2016

Senior Software Engineer · Hindustan Times

Configuration-driven RSS delivery for external clients (secure Basic-Auth endpoints, dynamic feeds) and a Spring Boot console app that parses and splits RSS feeds into per-category scheduled feeds with per-feed limits.

Stack: Vue.js · Quasar · Spring Boot · JDOM · Scheduler · RSS

PUBLISHING AUTOMATION (THOMSON DIGITAL)

Adobe InDesign Auto-Pagination Workflow

2008 – 2010

Executive — R&D · Thomson Digital · Thomson Press (India) Ltd.

XML-driven end-to-end pagination automation for academic publishers (Elsevier, Pearson), achieving 80–90% automated typesetting via the InDesign JavaScript SDK.

- Java + XSLT + XPath transformation engine converting publisher-specific XML DTDs into InDesign-ready structures.
- InDesign JS SDK scripts for auto table & image placement, bookmarking, linking, XML-out, and index-page generation.
- QA automation scripts for XML validation and content-quality checks integrated into the production workflow.

Stack: *Java · XML · XSLT · XPath · Adobe InDesign JS SDK*

INDEPENDENT & PERSONAL PROJECTS

QuoteAssist — AI-Powered Lead-to-Quote PlatformJun 2026 – Present

Founder / Architect · Personal · quoteassist.mytechbytes.in

Multi-tenant agentic AI platform that ingests travel quote requests from Outlook or web forms, extracts structured data with LLM + confidence scoring, enforces corporate policies via RAG, and drafts intelligent email replies — with adaptive auto-send graduation.

- Multi-channel ingestion: Outlook MS Add-in (MS Graph API), Phoenix LiveView web forms, and extensible webhooks normalised through one Elixir/OTP pipeline.
- LLM extraction + confidence gate: free-form text parsed into structured fields with per-field confidence scores; low-confidence items flagged for human review.
- RAG-based policy enforcement: fare class, supplier mandates, and budget thresholds checked against a vector-indexed corporate rules store; breaches rejected or escalated.
- Adaptive learning feedback loop: agent corrections become labelled training signals, fine-tuning the model and graduating trusted patterns to auto-send (kill-switch on confidence drop).
- Deployed on OCI always-free tier (Ubuntu 24.04, Docker Compose, Caddy) with Jenkins CI and OCIR (ap-mumbai-1).

Stack: *Elixir · Phoenix LiveView · OTP · Oban · Python · FastAPI · PostgreSQL · React · MS Graph API · LLM / GenAI · RAG · Docker · Jenkins · OCI*

Centralised Kiosk Display & Ad ManagementEarlier

Developer · Personal / Client

Scheduled advertisement display on Linux-based kiosks with an ad-wallet — debit on successful play, refund on failure. Deployed at Rajasthan bus terminals and shopping malls.

Stack: *Vue.js · PHP · Laravel · MySQL · Redis*